Dynamic network analysis of multi-scale -omics data for protein function prediction

Aydin Wells[1], Siyu Yang[1], Khalique Newaz[2], and Tijana Milenkovic[1]
[1]University of Notre Dame, United States
[2]University of Hamburg, Germany

A protein's amino acid sequence folds into a 3D structure, which directs what other proteins the given protein may interact with to carry out cellular function(s). Hence, analyses of protein structures are critical for understanding protein functions. Unfortunately, the functions of many proteins remain unknown because of the limitations of current biotechnologies (i.e., wet lab experimental approaches) for determining protein function. So, computational approaches for linking proteins' structures to their functions are necessary.

Previously, our lab captured protein structures (i.e., quantified protein structural features) using network-based methods, and tested these methods in the context of protein structure classification (PSC) – a task that is closely related to protein function prediction. PSC is a supervised problem of assigning proteins into pre-defined structural (e.g., CATH or SCOPe) classes based on the proteins' sequence or 3D structural features. Traditionally, PSC approaches have relied on sequence-based protein features. Because proteins with low (high) sequence similarity can have high (low) 3D structural similarity, 3D-structural features offer complementary insights to sequence features in the task of PSC. Traditional 3D-structural features are extracted directly from 3D structures of proteins. Instead, one can first model a protein's 3D structure as a protein structure network (PSN), where nodes are amino acids and two nodes are joined by an edge when the corresponding amino acids are close enough to each other in the protein's 3D structure (Fig. 1(a)). Then, one can use PSN (i.e., network) features of proteins for PSC. Our lab hypothesized that modeling 3D structures of proteins as PSNs could help gain novel insights about protein folding because it enables the application of powerful network science approaches to study 3D protein structures. In fact, our lab demonstrated that using PSN features often proved to be more accurate and typically faster than using state-of-the-art sequence and non-network-based 3D structural approaches in the PSC task [1].

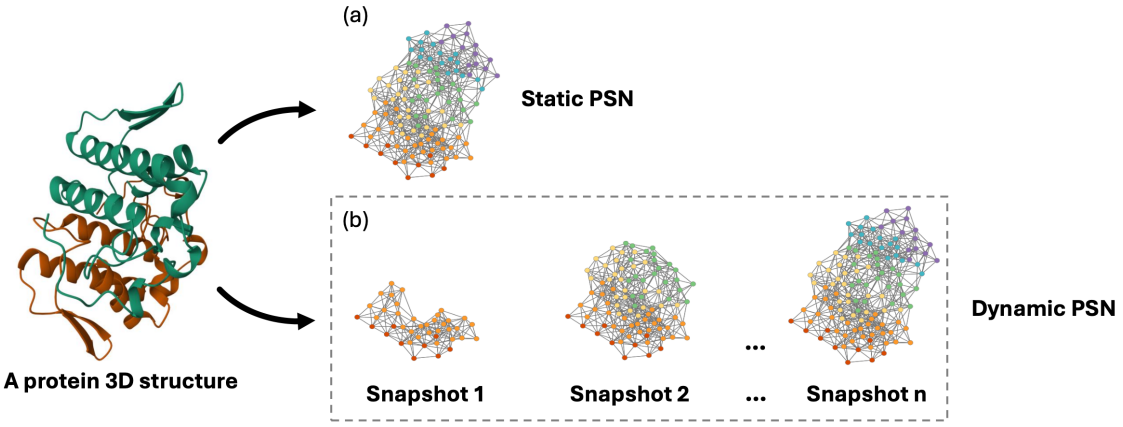

Figure 1: Illustration of a (a) static vs. (b) dynamic PSN for the protein with PDB ID 1AOK. The dynamic PSN consists of n network snapshots where the first snapshot (i.e, snapshot *1*) captures the initial portion of the protein sequence and its structure, and each next snapshot captures a larger portion of the protein sequence and its structure, while including all information from the previous snapshots. This continues until arriving at the last snapshot (i.e., snapshot n) which captures the full native structure.

However, traditional PSN methods (including our lab's previous approaches) have modeled the whole 3D structure of a protein as a static PSN, disregarding the temporal nature of the protein folding process (i.e., that some parts of the protein fold earlier than others before the protein acquires its native (final) 3D structure). Capturing such dynamic 3D structural information of a protein, which captures the 3D sub-structural (or intermediate) configurations of the protein as it undergoes folding to attain its native structure, can be more

informative. With this goal in mind, our lab recently proposed an idea of dynamic PSNs. Ideally, we would create dynamic PSNs using proteins' 3D sub-structures. However, there are very few kinds of experimental methods that can obtain such data, and such methods are not scalable. As a result, large-scale experimental data on protein 3D sub-structures are lacking. As a proxy, we did the best we could with the current data – we constructed dynamic PSNs using the native 3D structures of proteins, as follows.

A protein's native 3D structure is converted into a dynamic PSN by forming incremental cumulative groups of amino acids (i.e., sub-sequences) from the start to the end of the protein sequence. Then, a PSN snapshot is constructed for each incremental sub-sequence. All PSN snapshots collectively form a dynamic PSN (Fig. 1(b)). Despite the data limitation arising from lack of availability of dynamic protein 3D sub-structures, we demonstrated on 72 datasets (spanning 44,000 CATH and SCOPe protein domains) that modeling even native 3D structures of proteins as dynamic PSNs significantly improved upon the current state-of-the-art PSC approaches that are based on static PSNs [2] (and thus upon the existing state-of-the-art sequence and other 3D structural approaches that were already outperformed by static PSNs, per the above discussion).

We imagine that capturing dynamic 3D sub-structural data – rather than "just" the native structure – with dynamic PSNs will yield further improvements. But as stated before, such data are severally lacking. This is why in our previous study discussed above, our lab examined a dynamic model (i.e., dynamic PSNs) of static data (i.e., native protein 3D structures). Instead, in an on-going study, as a proxy to studying protein folding dynamics, we have identified large enough experimental data that capture how the structure of a protein dynamically changes before and after the protein is bound to a ligand (e.g., small molecule, ion, or another protein) – dynamic data that illustrate a dynamic process. Such structural changes that occur via the binding of a ligand are called ligand-bound protein motion types. When it binds to a protein, the ligand can induce structural rearrangements in the protein's structure, leading to changes in its overall shape, and as a result, can affect its function (Fig. 2(a)). Ligand-bound protein motion is dynamic; these types of motion are time-evolving processes. Given this dynamic data, currently, we are interested in investigating how effectively dynamic PSN features can capture dynamic information from the data, and whether they can do so better than modelling the data in a static fashion. More specifically, we have collected ligand-bound protein motion data from the Protein Structural Change DataBase (PSCDB), containing 3D structures of 1,084 protein-ligand pairs, where a pair corresponds to a protein before vs. after its binding to a ligand. Each protein-ligand pair is labeled with one of seven protein-ligand motion classes determined by wet-lab experimental results: coupled domain motion, independent domain motion, coupled local motion, independent local motion, burying ligand motion, no significant motion, and other motion. For each protein 3D structure before and after ligand binding, we model its structural change as a dynamic PSN using the same technique described before (Fig. 2(b)). We hypothesize that the dynamic PSN features extracted from the data will more accurately classify and explain the seven different types of protein structural changes than static PSNs. At the time of submitting this abstract, we are currently testing this hypothesis having consolidated these data and prepared our network-based approaches for studying them. We aim to present in our talk these currently unpublished results of this on-going analysis.

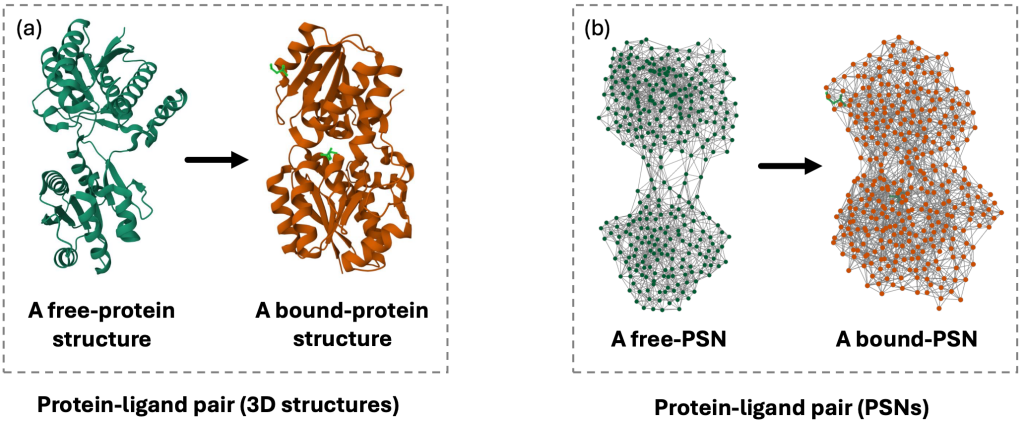

Figure 2: Illustration of protein structural change between (a) 3D structures and (b) the corresponding dynamic PSN capturing the protein's structural change before ("free") vs. after ("bound") binding to a ligand. The ligand is highlighted in neon green. This is an illustration for a protein with PDB IDs 1Z15 (*in dark green*) and 1Z17 (*in orange*)).

References

[1] K. Newaz, M. Ghalehnovi, A. Rahnama, P. J. Antsaklis, and T. Milenković, "Network-based protein structural classification," *Royal Society open science*, vol. 7, no. 6, 2020.

[2] K. Newaz, J. Piland, P. L. Clark, S. J. Emrich, J. Li, and T. Milenković, "Multi-layer sequential network analysis improves protein 3d structural classification," *Proteins: Structure, Function, and Bioinformatics*, vol. 90, no. 9, pp. 1721–1731, 2022.